

Research Paper

Personalized Course Recommendation System Using Learner Segmentation

Abstract

The rapid growth of online education platforms has transformed the learning ecosystem by making education accessible to learners across the world. However, the increasing number of courses available on such platforms creates difficulty for learners in identifying courses that align with their interests, skills, and career goals. Traditional recommendation systems often provide generic suggestions that fail to satisfy individual learner needs. This project proposes a Personalized Course Recommendation System using learner segmentation and data analytics techniques.

The project uses data collected from an online learning platform containing information about users, courses, instructors, and transactions. Exploratory Data Analysis (EDA) was performed to identify learner behavior patterns, course popularity, spending trends, and payment preferences. K-Means clustering was applied to segment learners into meaningful groups based on their engagement and spending patterns. Personalized recommendations were generated by identifying highly rated and popular courses within each learner segment.

A Streamlit dashboard was developed to visualize key analytics and recommendations interactively. The findings indicate that learner segmentation significantly improves recommendation relevance and helps platforms understand user behavior more effectively. The proposed system can increase learner engagement, improve course completion rates, and enhance overall platform revenue.

Chapter 1: Introduction

1.1 Background of the Study

The advancement of digital technology has significantly changed the education sector. Online learning platforms such as Coursera, Udemy, edX, and Skillshare provide learners with thousands of courses across multiple domains. These platforms generate massive amounts of learner interaction data including enrollments, ratings, payment transactions, and learning preferences.

Although online learning platforms provide flexibility and accessibility, learners often face difficulty selecting appropriate courses from a large catalog. Generic recommendation systems frequently fail to provide relevant suggestions because they do not consider individual learner behavior and interests.

Personalized recommendation systems use data analytics and machine learning techniques to analyze learner patterns and provide customized suggestions. Such systems improve learner satisfaction, engagement, and retention.

This project focuses on developing a Personalized Course Recommendation System using learner segmentation and data visualization techniques.

1.2 Problem Statement

Online learning platforms often recommend the same popular courses to all users without considering their learning behavior, interests, or spending patterns. As a result:

- Learners may lose interest in the platform.
- Course completion rates decrease.
- Platforms fail to maximize user engagement.
- Learners may not receive relevant learning opportunities.

Therefore, there is a need for an intelligent recommendation system capable of understanding learner behavior and generating personalized recommendations.

1.3 Objectives of the Study

The major objectives of this project are:

1. To analyze learner behavior using educational data.
 2. To perform Exploratory Data Analysis (EDA) on learner and course data.
 3. To identify patterns in learner enrollment and spending behavior.
 4. To segment learners using clustering techniques.
 5. To generate personalized course recommendations.
 6. To create an interactive dashboard for visualization.
 7. To provide strategic recommendations for educational platforms.
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1.4 Scope of the Study

The study focuses on analyzing learner interaction data from an online educational platform. The system can be used by:

- Online learning platforms

- Educational institutions
- Government skill-development programs
- Corporate learning platforms

The project demonstrates how machine learning and data analytics can support personalized education.

1.5 Significance of the Study

This project is significant because personalized learning is becoming increasingly important in modern education systems. The recommendation system developed in this project helps:

- Improve learner engagement
- Increase platform revenue
- Enhance course discoverability
- Support data-driven decision making
- Promote learner-centered education

The project also demonstrates practical applications of machine learning and business intelligence tools in education technology.

Chapter 2: Literature Review

Personalized recommendation systems have become a major area of research in educational technology. Previous studies have shown that recommendation systems can significantly improve learner satisfaction and course completion rates.

Collaborative filtering and content-based recommendation systems are commonly used approaches. However, these methods may not fully capture learner behavior patterns. Clustering techniques such as K-Means clustering help group learners based on similarities in behavior, spending, and interests.

Data visualization tools such as Power BI and Streamlit are increasingly used to present analytical insights interactively. Modern educational platforms combine machine learning, analytics, and visualization tools to improve learning experiences.

This project integrates clustering techniques with interactive dashboards to create an intelligent recommendation framework.

Chapter 3: Dataset Description

3.1 Dataset Overview

The dataset used in this project contains four major tables:

1. Users
2. Courses
3. Teachers
4. Transactions

These tables were merged to create a comprehensive learner profile dataset.

3.2 Users Table

The Users table contains demographic details of learners.

Attributes:

- UserID
- Name
- Age
- Gender
- Email

This table helps analyze learner demographics and preferences.

3.3 Courses Table

The Courses table contains course-related information.

Attributes:

- CourseID
- CourseName
- CourseCategory
- CoursePrice
- CourseRating
- CourseType

This table helps analyze course popularity and ratings.

3.4 Teachers Table

The Teachers table contains instructor information.

Attributes:

- TeacherID
- TeacherName
- Experience

This table helps analyze instructor performance.

3.5 Transactions Table

The Transactions table contains learner enrollment records.

Attributes:

- TransactionID
- UserID
- CourseID
- Amount
- PaymentMethod
- TransactionDate

This table helps analyze revenue and learner activity.

Chapter 4: Data Preprocessing

4.1 Data Cleaning

Data cleaning was performed to improve dataset quality. The following steps were performed:

- Removed duplicate records
- Checked missing values
- Standardized column names
- Verified data consistency

4.2 Data Integration

The four tables were merged using common keys:

- UserID
- CourseID
- TeacherID

This created a unified dataset for analysis.

4.3 Handling Missing Values

Missing values were handled using:

- Mean replacement for numerical fields
- “Unknown” labels for categorical fields

4.4 Feature Engineering

Additional features were created:

- Total learner spending
- Enrollment frequency
- Average course ratings
- Preferred categories

These features improved clustering accuracy.

4.5 Data Normalization

Numerical variables were normalized to ensure equal contribution during clustering.

The normalization formula used was:

$$x_norm = (x - x_min) / (x_max - x_min)$$

Chapter 5: Exploratory Data Analysis (EDA)

5.1 Revenue Analysis

Revenue analysis showed that premium and technical courses generated the highest revenue. Data Science and Programming categories contributed significantly to total platform earnings.

5.2 Course Popularity Analysis

Enrollment analysis revealed that beginner-friendly programming and AI-related courses had the highest enrollment counts.

5.3 Learner Demographics

The majority of learners belonged to the age group of 20–35 years. Gender distribution analysis showed balanced participation across categories.

5.4 Payment Method Analysis

Digital payment methods such as credit cards and UPI were the most commonly used payment options.

5.5 Rating Analysis

Highly rated courses received more enrollments, indicating that course quality strongly influences learner behavior.

Chapter 6: Learner Segmentation

6.1 Introduction to Clustering

Clustering is an unsupervised machine learning technique used to group similar data points. K-Means clustering was used in this project to identify learner segments.

6.2 Feature Selection

The following features were selected:

- Total courses enrolled
- Total spending
- Average course rating

- Number of categories explored
 - Age
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6.3 K-Means Clustering

K-Means clustering grouped learners into four clusters.

Cluster 1: Casual Learners

- Low enrollments
- Low spending

Cluster 2: Premium Learners

- High spending
- Frequent enrollments

Cluster 3: Explorers

- Interested in multiple categories

Cluster 4: Specialized Learners

- Focused on one domain
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6.4 Cluster Evaluation

Silhouette Score was used to evaluate clustering quality. The clustering results showed meaningful learner segmentation.

Chapter 7: Recommendation System

7.1 Recommendation Logic

Recommendations were generated based on:

- Learner segment
 - Popular courses
 - Course ratings
 - Category similarity
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7.2 Personalized Recommendations

Each learner received customized course suggestions based on their cluster characteristics.

Example:

- Premium learners received advanced technical courses.
 - Casual learners received beginner-friendly recommendations.
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Chapter 8: Dashboard Development

A Streamlit dashboard was developed to visualize:

- Revenue analytics
- Learner analytics
- Course analytics
- Recommendation insights

The dashboard provides interactive filtering and visualization capabilities.

Chapter 9: Key Findings

1. Premium learners generate the highest revenue.
 2. Technical courses are the most popular.
 3. Course ratings strongly influence enrollments.
 4. Personalized recommendations improve engagement.
 5. Learners prefer digital payment methods.
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Chapter 10: Recommendations

1. Implement AI-based personalized recommendations.
 2. Offer discounts to low-engagement learners.
 3. Promote highly rated courses.
 4. Develop category-specific learning paths.
 5. Improve beginner-level content availability.
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Chapter 11: Conclusion

The project successfully demonstrates the application of machine learning and analytics in online education platforms. Learner segmentation helps platforms understand user behavior and provide personalized recommendations. The developed recommendation system and interactive dashboard improve decision-making and learner engagement.

The project highlights the importance of data-driven education systems and demonstrates how analytics can enhance personalized learning experiences.

Future Scope

Future improvements may include:

- Deep learning-based recommendation systems
 - Real-time recommendation engines
 - NLP-based course analysis
 - Integration with mobile learning applications
 - Predictive learner performance analysis
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